Integrated Business Operations Platform (IBOP)

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Introduction

- Many small and mid-scale businesses still rely on **manual processes or disconnected software** to manage operations such as **sales, accounting, HR, and inventory**, leading to inefficiencies and data silos.
- In real-world business environments, lack of integration results in **delayed decision-making, reduced operational visibility, and increased chances of human error**.
- This project, **Integrated Business Operations Platform (IBOP)**, is being developed in **collaboration with a real business (Shrinath Enterprises)** to address these challenges in a practical setting.
- The system focuses on building a **centralized, real-time platform** that integrates multiple business functions into a single interface, improving coordination and efficiency.
- It provides **live data tracking, role-based access, and streamlined workflows**, enabling better control over daily operations and informed decision-making.

Literature Review

Research Paper Review	Identified Gaps	Our project contribution
<u>A literature review on ERP implementation: Methodologies, module, software, and policy(2024):-</u> The paper conducts a literature review of 19 research papers on ERP implementation. It analyzes ERP systems based on implementation methodologies, software types (open-source vs closed-source), modules, and company size. Different ERP implementation approaches (SAP ASAP, Oracle OUM, Odoo method) are compared and discussed.	No experimental validation or real system No performance comparison between ERP systes Does not evaluate which methodology is best Lacks user experience / usability analysis No intelligent/automated ERP system (no AI/ML)	Provides overview of ERP implementation methodologies (SAP, Oracle, Odoo) Identifies differences in implementation across companies Covers multiple case studies (19 papers) We focus on Improving user experience and we add smart automation systems
<u>ERP ODOO IMPLEMENTATION IN SMALL RETAILERS:-</u>The study uses an ERP implementation approach with Business Process Re-engineering (BPR). It analyzes existing business processes (As-Is) and redesigns them into ERP-based processes (To-Be). A Fit/Gap analysis is used to compare manual and ERP-based processes.Implementation is carried out using Odoo ERP across modules like accounting, inventory, purchase, and POS.	No comparison with other ERP systems No security or data privacy discussion No real-time analytics or decision support No reliable analytics dashborad	Provides structured financial reports Integrates all departments into single database Improves stock tracking using barcode system We Add Comparable Analytics using visual charts

Literature Review

Research Paper Review	Identified Gaps	Our project contribution
ERP SYSTEMS ARCHITECTURE FOR THE MODERN AGE: A REVIEW OF THE STATE OF THE ART TECHNOLOGIES:-This paper follows a **systematic literature review approach**, analyzing multiple ERP architecture models and technologies from academic and industrial sources. It categorizes ERP architectures into different styles and evaluates their characteristics, advantages, and limitations.	This paper does not give any real-world testing or performance comparison and it also lacks in giving optimization techniques	This paper Provides **complete understanding of ERP architecture Introduces:SOA (modular services) Cloud ERP (cost-efficient) Web-based systems Helps in designing scalable ERP systems and choosing right architecture**
Implementation of Odoo-Based ERP in The Case Study of Micro, Small and Medium Enterprises (MSME) :- The paper follows a structured ERP implementation approach using Odoo. It includes business need analysis, GAP analysis, system design using UML, module configuration, data migration, testing (Black Box & UTAUT), and deployment on cloud VPS.	Performance metrics were not used to track sales and other records of employees	Provides **real-time data flow across modules End-to-End workflow for implementation Example : From order to payment Use performance metrics to track sales and other records for the employee**

Literature Review

Research Paper Review	Identified Gaps	Our project contribution
Modular Architecture in .NET Enterprise Systems: Patterns, Practices, and Evolution toward Scalable Systems:- The paper analyzes modular architecture patterns in enterprise systems, including layered architecture, composite applications, domain-driven design (DDD), and service-oriented architecture (SOA). It studies how systems evolve from monolithic to modular structures.	This paper does not take consideration of the business workflow integration for the project and is only a theoretical paper with no real implementation	Layered architecture structure (UI / logic / database) **Modular feature system** Where Sales, Inventory, Accounts are separate modules
Open-source ERP software;-The paper provides a descriptive overview of various open-source ERP systems, comparing their features, scalability, and application areas.	This paper is practically jusst a listing of the existing tools and having a comprehensive view of them, there is no real implementation or proper analysis	Customization flexibility ensures that businesses can adapt the system to their specific needs. The system is also designed as a browser-based ERP, similar to WebERP and Openbravo, allowing access through a web browser.

Literature Review

Research Paper Review	Identified Gaps	Our project contribution
Design and Implementation of Inventory Management System for University :-The system is developed using a web-based architecture with PHP, SQL, and frontend technologies. It follows a structured design including system architecture, module division, and database design. The Waterfall model is used for development.	Only an Inventory management system not a integrated ERP System	**Clear workflow design** → Login → Order → Distribution → Report **Admin and User module separation** Which gives us a Role-based system **Auto generation** of **reports on a** Daily/weekly bases
Proprietary vs. Open-Source ERP: The Case of SAP and Odoo:-The paper performs a comparative analysis between proprietary ERP (SAP) and open-source ERP (Odoo) based on multiple factors such as cost, scalability, customization, and business suitability.	This study doesnt have a performance comparison of the ERP tools	**Use of Customization-first design** Which allows flexibility in workflows **Decision-based ERP design** So that system can adapt to the business size Shows real-world ERP selection factors

Literature Review

Research Paper Review	Identified Gaps	Our project contribution
Small-Medium Business Enterprise Management Application:-The system is developed using a structured software development lifecycle including requirement analysis, system design, development, testing, and deployment. It uses a cloud-based three-tier architecture with cross-platform support.	Security concerns were not merasured with minimal analytics.	An intricate system where all SEM systems can be worked on one application like sales, inventory and accounts Use of a simple UI model
Web – Based Intelligent Inventory Management System:-The system is developed using a web-based distributed architecture with a client-server model. It integrates an intelligent system using fuzzy logic to manage inventory across multiple stores and determine reorder levels. Algorithm Used:Fuzzy Logic (Mamdani Inference System)	Not beginner friendly and way too complex due to the use of Fuzzy logic with limited scalability proof.	Use of Ai for intelligent decisions and systems. Graphical representation of data with clear visuals and a dashboard

Problem Statement

Businesses, especially small and medium enterprises, often rely on multiple disconnected systems to manage sales, inventory, finance, and human resources. This leads to data inconsistency, manual and inefficient workflows, making it difficult to obtain real-time insights and make informed decisions.

This project aims to develop an Integrated Business Operations Platform (IBOP) that consolidates key business functions such as CRM, sales, inventory, finance, and HR into a centralized web-based system, improving operational efficiency and decision-making.

Objectives

1. To design and develop a centralized platform that integrates multiple business operations including sales, inventory, finance, and human resources.
2. To automate key workflows such as order processing, inventory updates, invoicing, and reporting to reduce manual effort and errors.
3. To implement a role-based access system (Admin/User) for secure and controlled data management.
4. To provide real-time data tracking and automated report generation for better business insights and decision-making.
5. To build a scalable and customizable system that can adapt to different business requirements.

Objective 1- IBOP

Implementation

1. Data Management

- Centralized state using **React Context API**
- Manages Products (Inventory) and Sales data
- Enables real-time synchronization across modules

2. Inventory Module

- Add and manage products (Name, Price, Stock)
- Display products in tabular format
- Supports dynamic product selection in sales

3. Sales Module

- Create sales with:
  - Company, Product, Quantity, Rate, Discount
- Auto-calculates:
  - Total = Quantity × Rate
  - Final Amount after discount

4. Business Logic Integration

- Sales linked with inventory
- Automatic stock reduction after sale
- Validates stock availability before transaction

5. UI & Navigation

- Modular dashboard (ERP-style interface)
- Routing using **React Router**
- Responsive UI using **Tailwind CSS**

6. Stem Architecture

- Modular frontend (Dashboard, Sales, Inventory)
- Separation of UI and logic (Context-based)
- Scalable for backend and multi-user support

System Design and Technologies Used

System Components

Frontend (React + Tailwind)

- Built using **React (Vite)** for fast UI development
- Styled with **Tailwind CSS** for responsive design
- Component-based architecture for modularity

State Management (Context API)

- Centralized data using **React Context API**
- Manages:
  - Products (Inventory)
  - Sales records

Routing (React Router)

- Implemented navigation between modules:
  - Dashboard
  - Sales
  - Inventory

Data Handling

- Uses local state (no backend for now)
- Designed for easy integration with Node.js backend

Core Logic & Feature

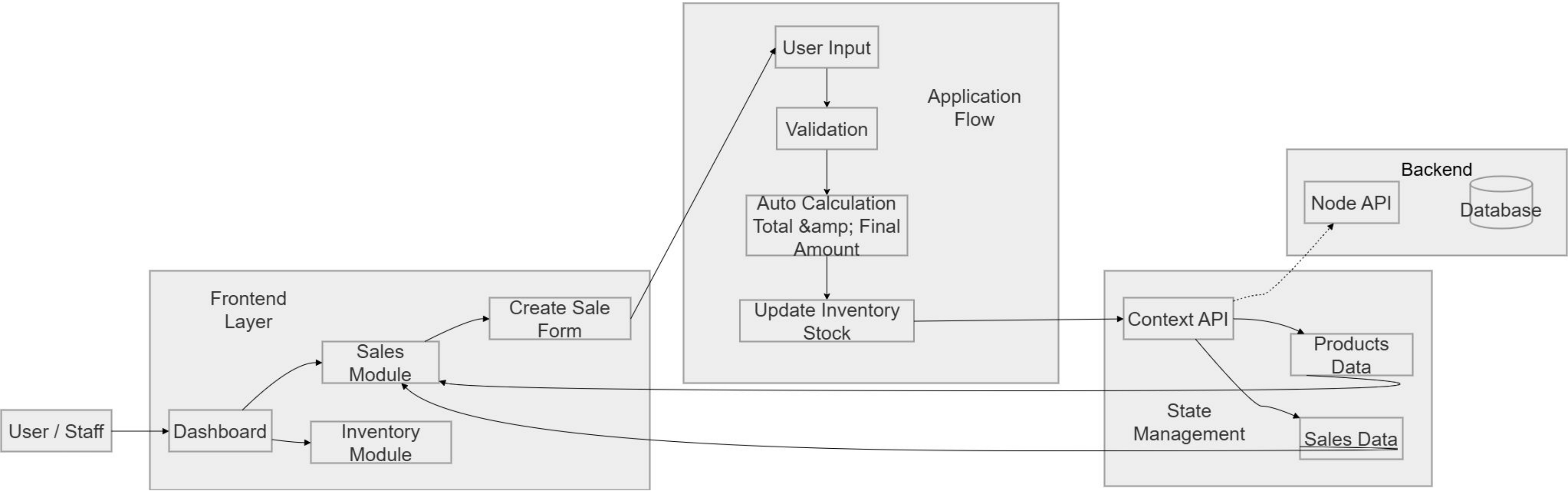
Key Functional Features

- **Sales–Inventory Integration:**
  - Sales automatically reduce product stock
- **Dynamic Calculations:**
  - Auto computes total and final amount
- **Validation Handling:**
  - Prevents invalid sales (e.g., insufficient stock)

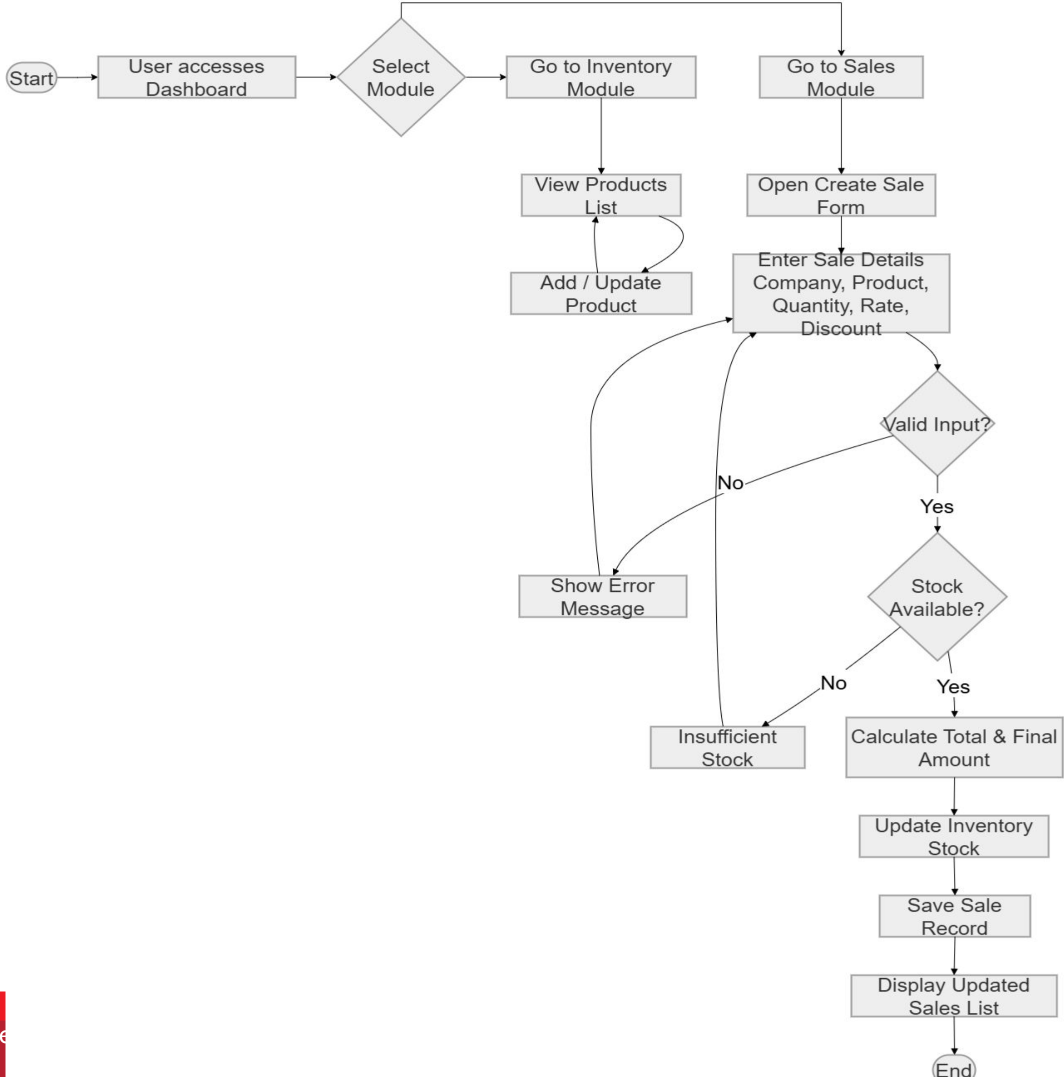
Project Advantages

- Real-time business operation tracking
- Simple and user-friendly interface
- Designed for real-world deployment in a pharma distribution business

Architecture Diagram

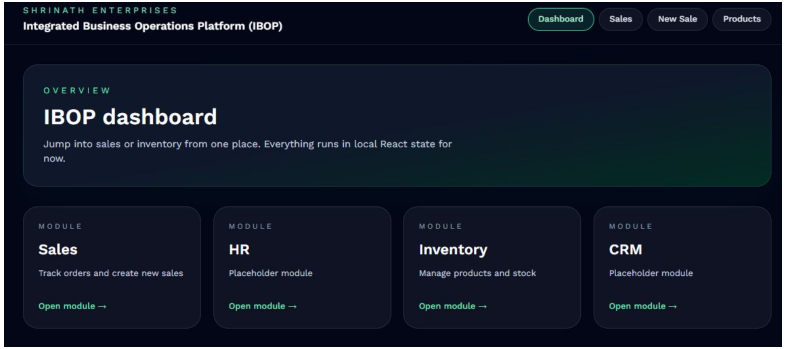


Flowchart Diagram For IBOP

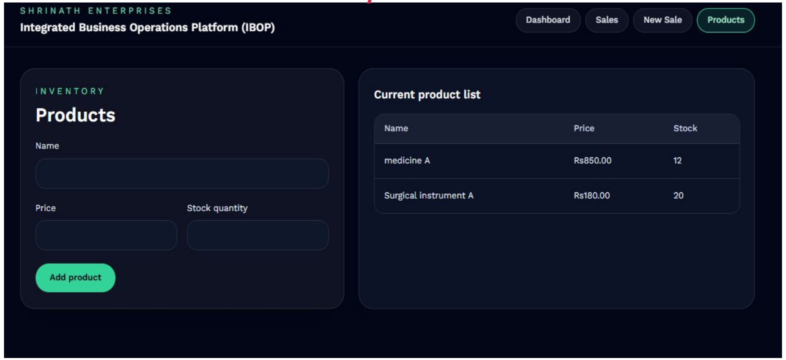


Implementation

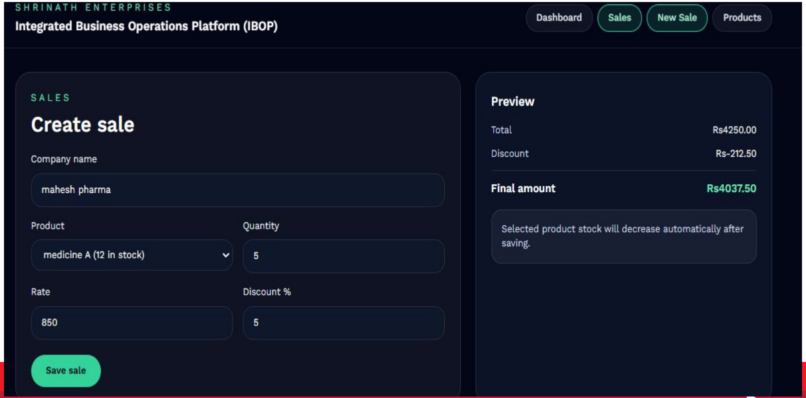
Modular Dashboard



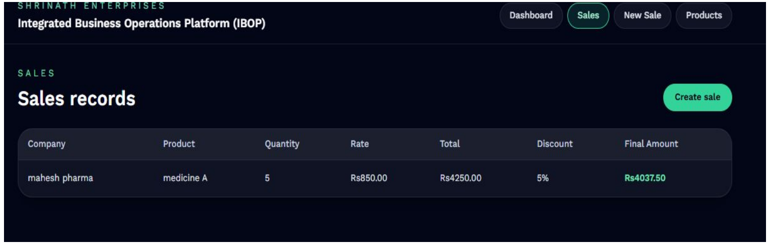
Inventory



Create a sale



Sales records



Conclusion

Developed a functional prototype of an Integrated Business Operations Platform (IBOP) that consolidates key business processes such as sales and inventory into a single system. The platform implements a modular ERP-like structure, allowing different components to interact seamlessly while maintaining a clean and scalable architecture.

I developed core functionalities including product management, sales creation, automatic stock updates, and real-time calculations for transactions. The system ensures data consistency by linking sales directly with inventory, reducing manual intervention and minimizing errors.

The application was built using modern frontend technologies with a focus on simplicity, usability, and real-world applicability. Features such as centralized state management, role-based workflow design, and a modular dashboard demonstrate how business operations can be digitized effectively.

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THANK YOU